

Q1. Define Covariance and explain how it differs from Correlation in terms of scale and interpretation.

Covariance is a statistical measure that indicates the direction of the linear relationship between two variables. It shows whether the variables increase together or one increases while the other decreases.

- Positive covariance: Both variables tend to increase or decrease together
- Negative covariance: One variable increases while the other decreases
- Zero covariance: No linear relationship

Formula :-

$$\text{Cov}(X, Y) = \frac{\sum(X - \bar{X})(Y - \bar{Y})}{n - 1}$$

Basis	Covariance	Correlation
Scale	Depends on the units of X and Y	Unit-free (standardized)
Range	No fixed range	Lies between -1 and +1
Interpretation	Shows direction only, magnitude is hard to interpret	Shows both direction and strength clearly
Units	Units are the product of units of X and Y	No units
Comparability	Cannot compare across datasets	Can be compared across datasets

Q2. What does a positive, negative, and zero covariance indicate about the relationship between two variables?

1. Positive Covariance

- Indicates that the two variables move in the same direction.
- When one variable increases, the other also tends to increase (and vice versa).

Example: Height and weight often show positive covariance.

2. Negative Covariance

- Indicates that the two variables move in opposite directions.
- When one variable increases, the other tends to decrease.

Example: Speed and travel time usually have negative covariance.

3. Zero Covariance

- Indicates no linear relationship between the variables.
- Changes in one variable do not systematically relate to changes in the other.

Q3. Discuss the limitations of covariance as a measure of relationship between two variables. Why is correlation preferred in many cases?

Limitations of Covariance

1. Depends on Scale and Units

- Covariance is affected by the units of measurement of the variables.
- Changing units (e.g., meters to centimeters) changes the covariance value, making it hard to interpret.

2. No Standard Range

- Covariance has no fixed minimum or maximum values.
- Because of this, it is difficult to judge the strength of the relationship.

3. Difficult Interpretation

- The magnitude of covariance does not clearly indicate how strong the relationship is.
- Only the sign (positive or negative) is meaningful.

4. Not Comparable Across Datasets

- Covariance values from different datasets cannot be directly compared due to scale dependency.

5. Only Measures Linear Relationship

- Covariance detects only linear relationships and fails to capture non-linear associations.

Correlation Is Preferred

1. Standardized Measure

- Correlation is a normalized form of covariance and is **unit-free**.

2. Fixed Range (-1 to +1)

- Makes it easy to interpret both **direction and strength** of the relationship.

3. Easy Comparison

- Correlation values can be compared across different datasets and variables.

4. Clear Interpretation

- Values close to +1 or -1 indicate strong relationships, while values near 0 indicate weak relationships.

Q4. Explain the difference between Pearson's correlation coefficient and Spearman's rank correlation coefficient. When would you prefer to use Spearman's correlation?

Aspect	Pearson's Correlation (r)	Spearman's Rank Correlation (ρ)
Type of data	Continuous, numerical data	Ordinal data or ranked data
Basis	Actual values of variables	Ranks of the data
Relationship measured	Linear relationship	Monotonic relationship (linear or non-linear)
Effect of outliers	Highly sensitive to outliers	Less sensitive to outliers
Distribution assumption	Assumes normally distributed variables	No normality assumption

Prefer Spearman's correlation when:

1. Data is ordinal or ranked

- Example: Student ranks, satisfaction levels.

2. Relationship is non-linear but monotonic

- Variables move consistently in one direction but not in a straight line.

3. Data contains outliers

- Spearman's correlation reduces the influence of extreme values.

4. Normality assumption is violated

- When data is skewed or not normally distributed.

Q5. If the correlation coefficient between two variables X and Y is 0.85, interpret this value in context. Can you infer causation from this value? Why or why not?

A correlation coefficient of 0.85 indicates a strong positive linear relationship between variables X and Y, meaning that as X increases, Y generally also increases. The value being close to +1 suggests a high degree of association between the two variables. However, this does not imply causation. Correlation only measures the strength and direction of a relationship, not a cause-and-effect link. The observed association could be due to the influence of a third variable, reverse causality, or even coincidence. Therefore, without additional evidence such as experimental or causal analysis, causation cannot be inferred from this correlation value.

Q6. Using the dataset below, calculate the covariance between X and Y.

X	2	4	6	8
Y	3	7	5	10

Mean

$$\bar{X} = \frac{2+4+6+8}{4} = \frac{20}{4} = 5$$

$$\bar{Y} = \frac{3 + 7 + 5 + 10}{4} = \frac{25}{4} = 6.25$$

X	Y	(X- $\bar{X}$)	(Y- $\bar{Y}$)	((X- $\bar{X}$)(Y- $\bar{Y}$))
2	3	-3	-3.25	9.75
4	7	-1	0.75	-0.75
6	5	1	-1.25	-1.25
8	10	3	3.75	11.25

$$\sum(X-\bar{X})(Y-\bar{Y})=9.75-0.75-1.25+11.25=19$$

Population Covariance

$$\text{Cov}(X, Y) = \frac{19}{4} = \boxed{4.75}$$

Sample Covariance

$$\text{Cov}(X, Y) = \frac{19}{3} \approx \boxed{6.33}$$

Q7. Compute the Pearson correlation coefficient between variables A and B:

A	10	20	30	40	50
B	8	14	18	24	28

Step 1: Find the means

$$\begin{aligned}\bar{A} &= \frac{10 + 20 + 30 + 40 + 50}{5} = 30 \\ \bar{B} &= \frac{8 + 14 + 18 + 24 + 28}{5} = 18.4\end{aligned}$$

A	B	A-30	B-18.4	(A-30)(B-18.4)	(A-30) ²	(B-18.4) ²
10	8	-20	-10.4	208	400	108.16
20	14	-10	-4.4	44	100	19.36
30	18	0	-0.4	0	0	0.16
40	24	10	5.6	56	100	31.36
50	28	20	9.6	192	400	92.16

Step 2: Compute deviations and products

$$\begin{aligned}\sum(A - \bar{A})(B - \bar{B}) &= 500 \\ \sum(A - \bar{A})^2 &= 1000 \\ \sum(B - \bar{B})^2 &= 251.2\end{aligned}$$

Step 3: Pearson correlation coefficient formula

$$r = \frac{\sum(A - \bar{A})(B - \bar{B})}{\sqrt{\sum(A - \bar{A})^2 \sum(B - \bar{B})^2}}$$

$$r = \frac{500}{\sqrt{1000 \times 251.2}}$$

$$r = \frac{500}{501.2} \approx 0.998$$

Q8. The following table shows heights (in cm) and weights (in kg) of 5 students. Find the correlation coefficient between Height and Weight.

Height	150	160	165	170	180
Weight	50	55	58	62	70

Step 2: Find the mean of Height and Weight

Mean of Height

$$\bar{X} = \frac{150 + 160 + 165 + 170 + 180}{5} = \frac{825}{5} = 165$$

Mean of Weight

$$\bar{Y} = \frac{50 + 55 + 58 + 62 + 70}{5} = \frac{295}{5} = 59$$

Step 3: Create the calculation table

X	Y	X-165	Y-59	(X-165)(Y-59)	(X-165) ²	(Y-59) ²
150	50	-15	-9	135	225	81
160	55	-5	-4	20	25	16
165	58	0	-1	0	0	1
170	62	5	3	15	25	9
180	70	15	11	165	225	121

Step 4: Find the sums

$$\sum(X - \bar{X})(Y - \bar{Y}) = 135 + 20 + 0 + 15 + 165 = 335$$

$$\sum(X - \bar{X})^2 = 225 + 25 + 0 + 25 + 225 = 500$$

$$\sum(Y - \bar{Y})^2 = 81 + 16 + 1 + 9 + 121 = 228$$

Step 5: Substitute values into the formula

$$r = \frac{335}{\sqrt{500 \times 228}}$$

$$r = \frac{335}{\sqrt{114000}}$$

$$r = \frac{335}{337.64}$$

$$r \approx 0.992$$

Q9. Given the dataset below, determine whether there is a positive or negative correlation between X and Y. ^[L]_{SEP} (No need for exact calculation, just reasoning.)

X	1	2	3	4	5
Y	15	12	9	7	3

Step 1: Observe the values of X

- X increases step by step: $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5$

Step 2: Observe the values of Y

- Y decreases step by step: $15 \rightarrow 12 \rightarrow 9 \rightarrow 7 \rightarrow 3$

Step 3: Compare the direction of change

- When X increases, Y decreases
- The variables move in opposite directions

Step 4: Apply the correlation rule

- Same direction (both increase or both decrease) $\rightarrow$ Positive correlation
- Opposite direction (one increases, other decreases) $\rightarrow$ Negative correlation

There is a *negative correlation* between X and Y.

Q10. Two investment portfolios have the following returns (%) over 5 years. Compute the covariance and correlation coefficient, and interpret whether the portfolios move together.

Year	Portfolio A	Portfolio B
1	8	6
2	10	9
3	12	11
4	9	8
5	11	10

Step 1: Find the mean (average) returns

Mean of Portfolio A

$$\bar{X} = \frac{8 + 10 + 12 + 9 + 11}{5} = \frac{50}{5} = 10$$

Mean of Portfolio B

$$\bar{Y} = \frac{6 + 9 + 11 + 8 + 10}{5} = \frac{44}{5} = 8.8$$

X	Y	X-10	Y-8.8	(X-10)(Y-8.8)	(X-10) ²	(Y-8.8) ²
8	6	-2	-2.8	5.6	4	7.84
10	9	0	0.2	0	0	0.04
12	11	2	2.2	4.4	4	4.84
9	8	-1	-0.8	0.8	1	0.64
11	10	1	1.2	1.2	1	1.44

Step 3: Find required sums

$$\sum(X - \bar{X})(Y - \bar{Y}) = 5.6 + 0 + 4.4 + 0.8 + 1.2 = 12$$

$$\sum(X - \bar{X})^2 = 4 + 0 + 4 + 1 + 1 = 10$$

$$\sum(Y - \bar{Y})^2 = 7.84 + 0.04 + 4.84 + 0.64 + 1.44 = 14.8$$

Step 4: Compute the covariance

Formula (using population covariance):

$$\begin{aligned}\text{Cov}(X, Y) &= \frac{\sum(X - \bar{X})(Y - \bar{Y})}{n} \\ \text{Cov}(X, Y) &= \frac{12}{5} = 2.4\end{aligned}$$

Step 5: Compute the correlation coefficient

$$\begin{aligned}r &= \frac{\sum(X - \bar{X})(Y - \bar{Y})}{\sqrt{\sum(X - \bar{X})^2 \sum(Y - \bar{Y})^2}} \\ r &= \frac{12}{\sqrt{10 \times 14.8}} \\ r &= \frac{12}{\sqrt{148}} \\ r &= \frac{12}{12.17} \\ r &\approx 0.99\end{aligned}$$